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1. **Experiment:** Transduction of sgRNA and Directed Evolution Screening of Raji Cells
 2. **Time:** 2026.04.19-2026.04.30
 3. **Member:** Kaiqing Zhang, Binxuan Zhang, Qichang Liu, Ziyang Yu, Yue Chao, Xinru Yuan
 4. **Materials:** Hek293T cells, Raji cells stably expressing dCas9-AID* Δ (or co-transduction with pCDH-dCas9-AID* Δ), pLX-sgRNA transfer plasmid (containing U6-sgRNA scaffold and BSD resistance gene), pMD2.G envelope plasmid, pCMVR8.74 packaging plasmid, Hieff Trans® Polyethylenimine Linear (PEI), DMEM high glucose, RPMI-1640 medium, 10% FBS, 1% penicillin-streptomycin, Polybrene, Blasticidin, antibiotic-free medium, recombinant human IL-6, recombinant human sIL-6R, EZ-Link™ Sulfo-NHS-Biotinylation Kit (Thermo Scientific, Cat. No. 21425), Zeba Spin desalting columns, Alexa Fluor 488 Streptavidin (Invitrogen, Cat. S11223), Anti-Alexa Fluor 488 polyclonal antibody (Invitrogen, Cat. A-11094), glycine (pH 2.5), 40 μ m cell strainers, flow cytometer and cell sorter (FACS Aria III), FACS buffer (PBS with 1% FBS), 0.45 μ m syringe filters, 100 mm culture dishes, 6-well plates, 15 mL conical tubes, centrifuge, biosafety cabinet, automated cell counter.
 5. **Method:**
 - (1) Preparation of Hek293T packaging cells:
 - ① Hek293T cells were cultured in 10 cm dishes at 37 °C in a 5% CO₂ incubator.
 - ② When cells reached approximately 75% confluence, the medium was replaced with 10.5 mL pre-warmed antibiotic-free medium (5% FBS) 2-4 h before transfection.
 - (2) Construction of pLX-sgRNA expressing the target sgRNA
 - ① Primer design and PCR amplification: Primers were designed to flank the NheI and NdeI sites on the synthesized fragment. The fragment was amplified by PCR using a high-fidelity DNA polymerase. The PCR product was analyzed by agarose gel electrophoresis, purified using a gel extraction kit, and quantified.
 - ② Double digestion: The purified PCR product and the pLX-sgRNA vector were separately digested with NheI and NdeI at 37 °C for 1-2 h. The digested products were purified using a PCR purification kit.
 - ③ Ligation: The digested PCR fragment and the digested pLX-sgRNA vector were ligated at a molar ratio of 3:1 using T4 DNA ligase at 16 °C overnight.
 - ④ Transformation and colony screening: The ligation mixture was transformed into competent *E. coli* cells. Colonies were selected on LB agar plates containing ampicillin. Single colonies were picked and cultured overnight in LB medium containing ampicillin. Plasmid DNA was extracted and verified by restriction enzyme digestion and Sanger sequencing using primers flanking the U6 promoter and gRNA scaffold.
 - ⑤ Sequence confirmation: Correct clones containing the target sgRNA sequence were confirmed by Sanger sequencing. The verified plasmid was named pLX-sgRNA-target and used for subsequent lentiviral packaging.

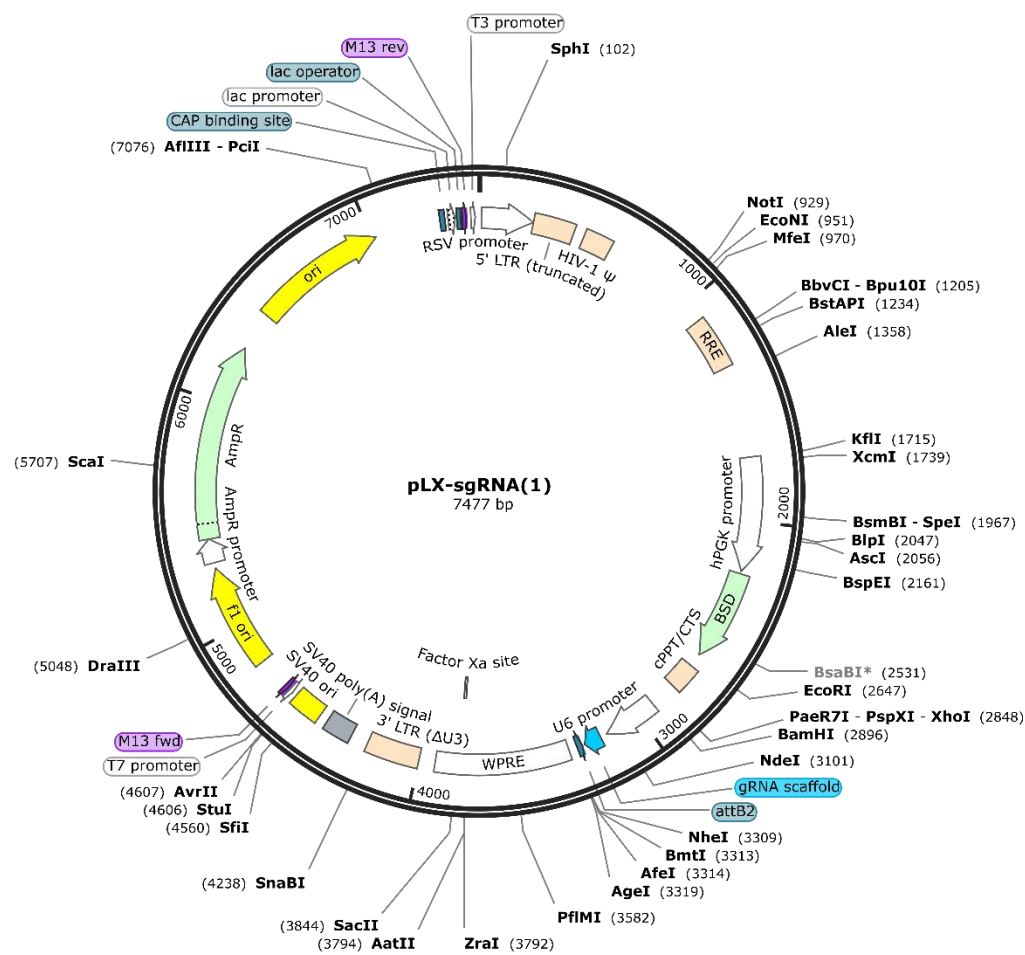


Figure 1. Map of the pLX-sgRNA lentiviral vector for sgRNA expression.

Table 1. The Sequence of sgRNA.

sgRNA	Sequence
A2-sgRNA-116 forw	GGCTAAGAAGAACCTTCCCG
B3-sgRNA-101 forw	GACTCCGAGCGATCTCGAGT
C3-sgRNA-53 forw	GGACCCGGTCGAAATTAGGC

(3) Lentiviral packaging of pLX-sgRNA-target:

- ① The transfer plasmid pLX-sgRNA-target, envelope plasmid pMD2.G, and packaging plasmid pCMVR8.74 were mixed at a mass ratio of 3:1:2, with a total DNA amount of 30 µg per 10 cm dish. Transfection was performed using Hieff Trans® PEI at a PEI:DNA mass ratio of 3:1. The transfection mixture composition per 10 cm dish is shown in Table 2.

Table 2. Transfection mixture composition.

Transfection tube	Reagent	Volume/Amount
A	PEI (1 mg/mL)	90 µL
	Antibiotic-free medium	910 µL (to 1 mL total)
	pLX-sgRNA-target plasmid	15 µg
	Envelope plasmid (pMD2.G)	5 µg
B	Packaging plasmid (pCMVR8.74)	10 µg
	Antibiotic-free medium	Up to 1 mL

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- ② The PEI solution (Tube A) and the plasmid mixture (Tube B) were gently mixed separately, then combined by adding Tube B dropwise into Tube A. The mixture was vortexed briefly and incubated at room temperature for 15 min.
 - ③ The transfection complexes were added dropwise to the Hek293T cells, and the plates were returned to the incubator.
 - ④ 6 h after transfection, the medium was replaced with fresh complete culture medium.
- (4) Virus supernatant collection and concentration:
- ① At 48 h and 72 h post-transfection, viral supernatants were collected from Hek293T cells, filtered through a 0.45 μm filter, and used immediately or stored at $-80\text{ }^{\circ}\text{C}$.
 - ② The filtered viral supernatant was concentrated by ultracentrifugation at $20,000 \times g$ for 99 min at $4\text{ }^{\circ}\text{C}$. The viral pellet was resuspended in PBS or complete culture medium, aliquoted, and stored at $-80\text{ }^{\circ}\text{C}$ until use.
- (5) Transduction of Raji cells expressing dCas9-AID* Δ :
- ① Raji cells stably expressing dCas9-AID* Δ were collected from a culture flask and centrifuged at $300 \times g$ for 5 min at room temperature.
 - ② The supernatant was removed, and the cell pellet was resuspended in complete medium (RPMI-1640 with 10% FBS and 1% penicillin-streptomycin).
 - ③ Cell density was determined, and the suspension was diluted to 1×10^6 cells/mL. For each well of a 6-well plate, 2 mL of cell suspension (2×10^6 cells) was dispensed and incubated at $37\text{ }^{\circ}\text{C}$, 5% CO_2 for 24 h before infection.
 - ④ Infection: The concentrated lentiviral supernatant was added at a multiplicity of infection (MOI) of 30. Polybrene was added to a final concentration of 8 $\mu\text{g/mL}$. Cells were subjected to spin infection at 1,500 rpm for 1.5 h at room temperature, then transferred to a $37\text{ }^{\circ}\text{C}$, 5% CO_2 incubator.
- (6) Co-culture and Directed Evolution Screening:
- ① 16 h after infection, the cell suspension was collected by centrifugation at $300 \times g$ for 5 min, the supernatant containing virus and Polybrene was discarded, and the cell pellet was resuspended in fresh complete medium (RPMI-1640 with 10% FBS and 1% penicillin-streptomycin).
 - ② 48 h after transduction, Blasticidin was added to the culture at a final concentration of 10 $\mu\text{g/mL}$. The working concentration was pre-determined by a kill curve on untransduced Raji cells to ensure complete killing of non-transduced cells within 5-7 days. The medium was replaced with fresh complete medium containing Blasticidin every 2-3 days.
 - ③ After selection, the surviving cells were expanded, and the AF488-labeled IL-6/sIL-6R α complex, which had been constructed in a previous experiment, was added to initiate the directed evolution.
 - ④ The surviving cells were then expanded in complete medium and cultured for a total of 14 days after transduction, so that dCas9-AID* Δ -mediated mutagenesis of the target locus could accumulate. Cell density was determined daily, and the culture was passaged whenever the density exceeded 1×10^6 cells/mL to keep the cells in logarithmic growth.
 - ⑤ After 14 days of co-culture, the surviving Raji cells in the system were collected for subsequent flow cytometric detection and sorting. Meanwhile, an aliquot of cells was retained before sorting and cryopreserved as a backup.
- (7) Endocytosis assay and flow cytometric sorting of complex-internalizing cells:
- ① After 14 days of co-culture, the surviving Raji cells were collected by centrifugation at $300 \times g$ for 5 min at room temperature. The supernatant was discarded, and the cell pellet was resuspended in ice-cold PBS.
 - ② Acid wash to remove surface-bound complex: The cells were treated with 0.2 M glycine (pH 2.5) for 5 min at room temperature to remove AF488 complex bound to the cell surface but not internalized,

then neutralized with complete medium and washed twice with ice-cold PBS.

- ③ The cells were resuspended in FACS buffer (PBS with 1% FBS) at a density of 1×10^7 cells/mL and passed through a 40 μ m cell strainer to obtain a single-cell suspension.
- ④ Surface fluorescence quenching: Each sample was divided into two identical aliquots: Unquenched and Quenched. The Quenched aliquot was incubated with anti-Alexa Fluor 488 polyclonal antibody (Invitrogen, Cat. A-11094) at a final concentration of 10 μ g/mL on ice for 30 min in the dark. The Unquenched aliquot received an equal volume of FACS buffer. The antibody does not enter live cells and only quenches AF488 fluorescence on the cell surface.
- ⑤ Flow cytometric detection and sorting: Nozzle size 100 μ m; sort mode: population enrichment (bulk sorting). Gating strategy: single cells $\rightarrow$ live cells $\rightarrow$ Raji cells $\rightarrow$ intracellular AF488 fluorescence. The gating boundary was established using unstained cells, the 4 $^{\circ}$ C internalization-inhibited control, and the isotype control. Quenched AF488-positive cells were collected as cells that had successfully internalized the complex.
- ⑥ The sorted cells were collected into sterile tubes containing complete medium supplemented with 20% FBS and expanded at 37 $^{\circ}$ C in a 5% CO₂ incubator for subsequent genomic DNA extraction and third-generation sequencing.
- ⑦ The mean fluorescence intensity (MFI) and the percentage of AF488-positive cells were recorded for each group. The internalization efficiency of the sgRNA-transduced library was compared with that of untransduced Raji cells to evaluate the effect of sgRNA-guided mutagenesis on the uptake of the IL-6/sIL-6R α complex.

6. Result:

After 14 days of culture, the sgRNA-transduced library was incubated with the AF488-labeled IL-6/sIL-6R α complex, and internalization was analyzed by flow cytometry. Cells positive for intracellular AF488 fluorescence were gated against the 4 $^{\circ}$ C internalization-inhibited control, collected by bulk sorting, expanded, and processed for genomic DNA extraction and third-generation sequencing.

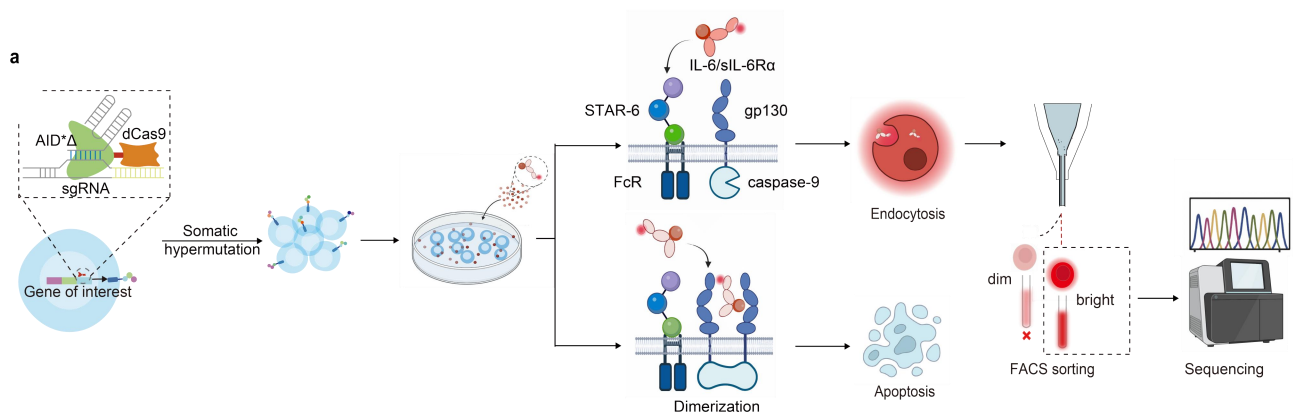


Figure 2. Schematic of the second-round directed-evolution and sequencing workflow. The A2, B3 and C3 segments were assembled into a continuous open reading frame, and the experimental group was established using sgRNAs targeting the A2, B3 or C3 region. Following fluorescence-based sorting with the IL-6/sIL-6R α complex, an amplicon spanning all three segments was subjected to PacBio HiFi full-length sequencing.